

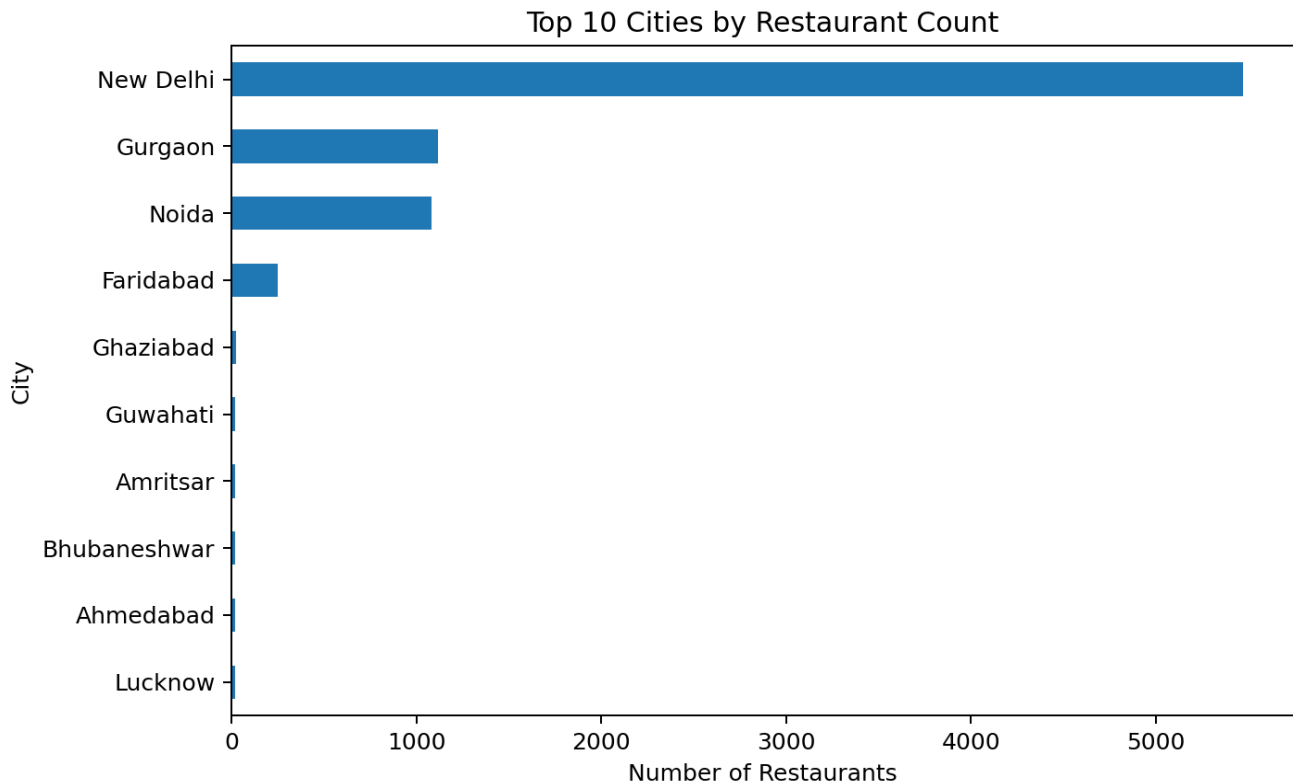
Cognifyz Technologies — Machine Learning Internship

Task 2: Restaurant Recommendation System

Objective: Create a restaurant recommendation system based on user preferences using a content-based filtering approach, with interpretable preference-aware ranking.

1. Dataset Overview

The supplied dataset contains 9,551 restaurants, 21 columns, 141 cities and 1,826 distinct Cuisine entries. The Cuisine field contains 9 missing values, which were mapped to 'Unknown'.



2. Recommendation Design

The system represents each restaurant using TF-IDF features from Cuisines, City and Locality. A user query is converted into the same vector space and cosine similarity measures content relevance. The ranking is then refined with price-range preference, restaurant rating, review volume (log-transformed votes), and optional table-booking/online-delivery preferences. This is intentionally more informative than an exact-match filter.

Ranking formula

Base score = $0.65 \times \text{content similarity} + 0.15 \times \text{price preference} + 0.10 \times \text{rating quality} + 0.05 \times \text{review-volume signal} + \text{optional service-preference bonuses}$.

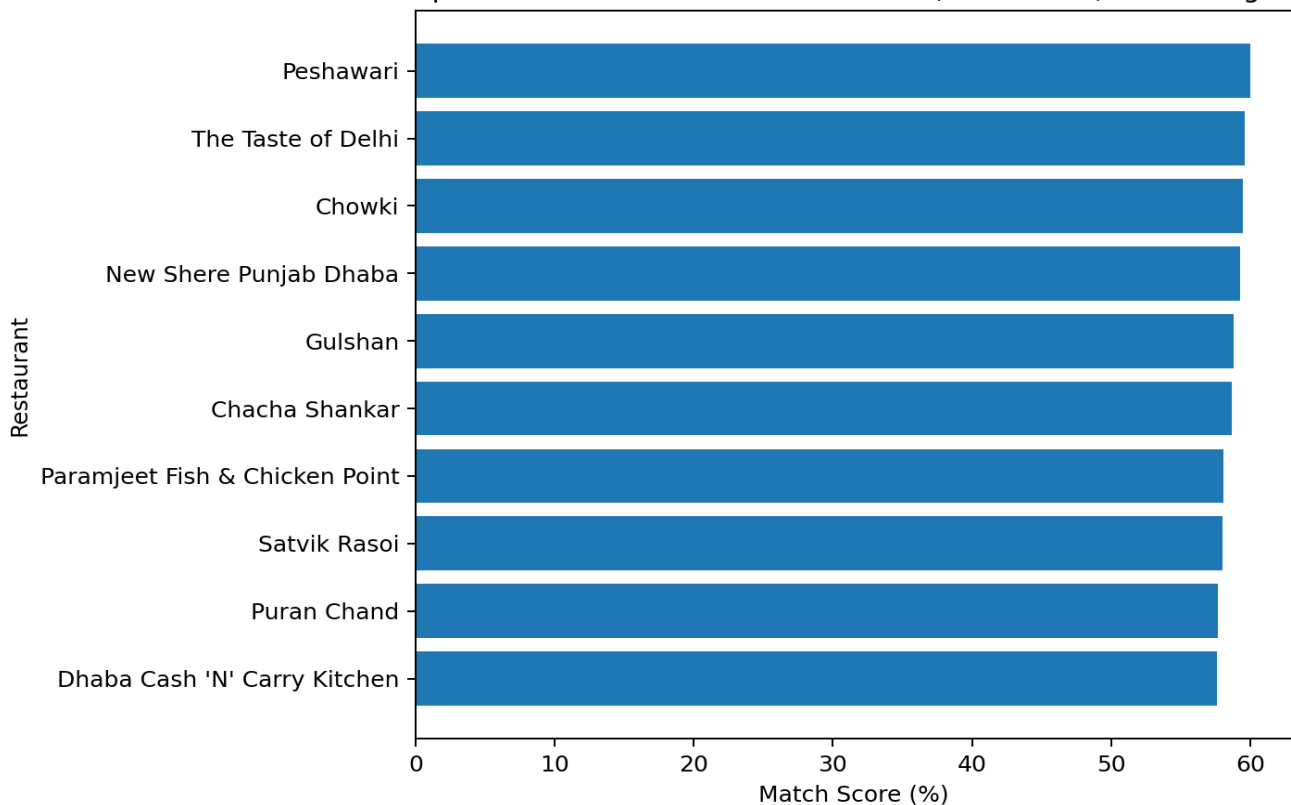
3. Sample User Test

Sample preference: **North Indian** cuisine, **New Delhi**, price range **2**. The top 10 recommendations have an average match score of **58.7%** and contain **1** distinct cuisine strings, demonstrating ranked rather than binary filtering.

Restaurant	City	Cuisine	Price	Rating	Match
Peshawari	New Delhi	North Indian	2	3.6	60.0%
The Taste of Delhi	New Delhi	North Indian	2	3.3	59.6%

Restaurant	City	Cuisine	Price	Rating	Match
Chowki	New Delhi	North Indian	2	3.7	59.5%
New Shere Punjab Dhaba	New Delhi	North Indian	2	3.6	59.3%
Gulshan	New Delhi	North Indian	2	3.3	58.8%
Chacha Shankar	New Delhi	North Indian	2	3.2	58.7%
Paramjeet Fish & Chicken Point	New Delhi	North Indian	2	3.1	58.1%
Satvik Rasoi	New Delhi	North Indian	2	3.8	58.0%
Puran Chand	New Delhi	North Indian	2	2.6	57.7%
Dhaba Cash 'N' Carry Kitchen	New Delhi	North Indian	2	3.1	57.6%

Top Recommendations — North Indian, New Delhi, Price Range 2



4. Engineering Considerations

- High-cardinality categorical data is represented with TF-IDF rather than one-hot encoding every unique cuisine/location string.
- The system supports partial preferences: a user can provide cuisine, city, locality, price range and optional service preferences.
- Recommendations are ranked by a continuous score, allowing the top results to be compared instead of returning an arbitrary matching list.
- Unrated restaurants are suppressed when enough rated alternatives exist, improving practical recommendation quality.
- The score is a ranking heuristic, not a probability of user satisfaction. In production, it should be calibrated and validated with user interaction data.

5. Future Improvements

A production version could incorporate user-item interaction history, learning-to-rank, semantic embeddings for cuisines, geographic distance, diversity constraints, and online evaluation metrics such as Precision@K,

Recall@K and NDCG@K.

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